PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Self-harm and suicidal behaviour in Ghana: a systematic review protocol
AUTHORS	Quarshie, Emmanuel; Oppong Asante, Kwaku; Andoh-Arthur, Johnny

VERSION 1 – REVIEW

REVIEWER	Pozuelo, Julia University of Oxford, Psychiatry
REVIEW RETURNED	27-Jul-2021

GENERAL COMMENTS	This study will address an important problem in an underserved population. Measuring self-harm and suicidal behaviour in Ghana is an essential starting point to understand the magnitude of the problem, inform policy, and guide interventions targeting mental health. The protocol is well-structured and clear. I have two main comments: 1. Defining and measuring outcomes. I would like to see a clearer explanation on the outcomes and how should these be measured. For example, will the authors consider studies that include both suicidal self-harm and non-suicidal self-harm? Regarding prevalence rates of self-harm and suicidal behaviours, will authors include studies that use self-report scales, diagnostic instruments, reports from family? Only validated measures vs. any available measure? 2. Study design. I found the study design categorization confusing. For example, the authors state that they will include longitudinal studies and cohort studies – but these are the same thing. I suggest they stick to standard study design categorization (see Figure 1 in Grimes and Schulz 2002 Lancet paper). Also, I wonder whether including such a wide range of study types is the most efficient way to answer the question – wouldn't it make more sense to focus on observational studies? Some smaller questions for the authors: 3. Types of study: What will authors do in cases where multiple studies use the same data? (i.e. select study with the larger sample size? Or the most recent publication date?) 4. Selection process: Who will run the search, compile results, and eliminate duplicates? A single researcher? If so, say so. Also, how will you resolve disagreement between both reviewers if it cannot be resolved by consensus? Will it be referred to a third party?

	5. Data collection process: Will two reviewers independently extract data from each article or will one reviewer extract, and then another reviewer will check for completion and accuracy? 6. Data synthesis: Will you only include studies that present data as odds ratio and adjust for confounding factors (age, sex, socioeconomic status? Or all types of data?

REVIEWER	Thippaiah, Srinagesh Hofstra Northwell School of Medicine at Hofstra University
REVIEW RETURNED	02-Aug-2021

GENERAL COMMENTS	1. It is good that authors clearly defined the self harm and suicidal behavior, as it various its meaning in different cultures and I am hoping that authors have took cultural variation in the meaning of self harm behavior in the context of Ghana. 2. In some countries suicide is punishable by law, if it so in Ghana or not ? Then, please make it clear in the protocol 3. It is commendable efforts by the authors to do this review to bring reformations in legislation of the country 4. The inclusion criteria seems to be too broad this might capture too many irrelevant papers 5. PICO frame work: Though it is not explicitly mentioned in the protocol but it appears they have followed PICO framework to develop the search strategy . It nice authors have used PRISMA guidelines 6. I could find the example of search terms that they are planning to use 7. In the page 7 line 31-37 authors says there is one national level study and 2 regional level study. How they are different from the authors proposal of the current review ?? How is your current proposed study is different and robust from the previous ones?? 8. The outcome measure should have more clearer justification why such specific measures were selected 9. I am not sure why the clinical based studies were excluded?? May be authors should explain how it causes the bias?? Does that not miss majority of important studies and population at risk ?? 10. Assessment of Bias and statistical measures were seeming to be appropriate. Because authors have decided to take wide variety of studies that combining such results going to be a very daunting task.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1: Dr Julia Pozuelo, University of Oxford
Comment 1: Defining and measuring outcomes. I would like to see a clearer explanation on the outcomes and how should these be measured. For example, will the authors consider studies that

include both suicidal self-harm and non-suicidal self-harm? Regarding prevalence rates of self-harm and suicidal behaviours, will authors include studies that use self-report scales, diagnostic instruments, reports from family? Only validated measures vs. any available measure?
Authors' response: We thank the reviewer for this helpful comment. However, we do not wish to draw a sharp distinction between suicidal and non-suicidal acts, as suggested by Dr Pozuelo. We have added to the text, explaining our rationale (pp.6-8).

Comment 2: Study design. I found the study design categorization confusing. For example, the authors state that they will include longitudinal studies and cohort studies – but these are the same thing. I suggest they stick to standard study design categorization (see Figure 1 in Grimes and Schulz 2002 Lancet paper). Also, I wonder whether including such a wide range of study types is the most efficient way to answer the question – wouldn't it make more sense to focus on observational studies?
Authors' response: Thank you, Dr Pozuelo, for this suggestion. We have now clarified that we will focus on observational studies (i.e., cross-sectional studies, case control studies, and cohort studies) to draw evidence to address our review's objectives regarding prevalence and associated factors. To address the review's objectives on reported reasons and means of self-harm and suicidal behaviour, we will draw evidence reported mainly by eligible qualitative studies (p.8).

Comment 3: Types of study: What will authors do in cases where multiple studies use the same data? (i.e. select study with the larger sample size? Or the most recent publication date?)

Authors' response: We have addressed this comment (p.8).

Comment 4: Selection process: Who will run the search, compile results, and eliminate duplicates? A single researcher? If so, say so. Also, how will you resolve disagreement between both reviewers if it cannot be resolved by consensus? Will it be referred to a third party?
Authors' response: We have addressed this comment: see "Search strategy and process" (p.9).

Comment 5: Data collection process: Will two reviewers independently extract data from each article or will one reviewer extract, and then another reviewer will check for completion and accuracy?

Authors' response: We have addressed this comment (p.10).

Comment 6: Data synthesis: Will you only include studies that present data as odds ratio and adjust for confounding factors (age, sex, socioeconomic status? Or all types of data?
Authors' response: We have addressed this comment (p.11).

Reviewer 2: Prof S. Thippaiah, Hofstra Northwell School of Medicine, Hofstra University
Comment 1: It is good that authors clearly defined the self harm and suicidal behavior, as it various its meaning in different cultures and I am hoping that authors have took cultural variation in the meaning of self harm behavior in the context of Ghana.
Authors' response: We thank Prof Thippaiah for this comment. Considering the lack of evidence on the contribution from Ghana to the contextual conceptualisation of self-harm, we have adopted the WHO definition, but also, we are mindful of the peculiarity of the criminality of attempted suicide in Ghana (p. 4, 7).

Comment 2: In some countries suicide is punishable by law, if it so in Ghana or not ? Then, please make it clear in the protocol.
Authors' response: We have now made this explicit (p.4).

Comment 3: It is commendable efforts by the authors to do this review to bring reformations in

legislation of the country.
Authors' response: Thank you.

Comment 4: The inclusion criteria seems to be too broad this might capture too many irrelevant papers
Authors' response: We have now streamlined our eligibility criteria to allow for more specificity (see Table 1) – pp.6-8

Comment 5: PICO frame work: Though it is not explicitly mentioned in the protocol but it appears they have followed PICO framework to develop the search strategy . It nice authors have used PRISMA guidelines
Authors' response: We have elected to be guided by the PRISMA guidelines, as this allows the review to also follow the PICO framework.

Comment 6: I could find the example of search terms that they are planning to use
Authors' response: We have addressed this comment (p.9).

Comment 7: In the page 7 line 31-37 authors says there is one national level study and 2 regional level study. How they are different from the authors proposal of the current review ?? How is your current proposed study is different and robust from the previous ones??
Authors' response: We have addressed this comment (p.4).

Comment 8: The outcome measure should have more clearer justification why such specific measures were selected

Authors' response: We have addressed this comment (pp.6-8).

Comment 9: I am not sure why the clinical based studies were excluded?? May be authors should explain how it causes the bias?? Does that not miss majority of important studies and population at risk ??

Authors' response: We have not totally excluded clinic-based studies from our proposed review. We have only specifically indicated one occasion on which we will exclude clinic-based studies, that is, with regard to prevalence studies, where we argued thus, "Clinic-based (accident and emergency/outpatient/ inpatient department) prevalence studies will be excluded due to the inherent bias in selection of patients and patients' access to clinical care that could skew results artificially." (p.8).

Comment 10: Assessment of Bias and statistical measures were seeming to be appropriate. Because authors have decided to take wide variety of studies that combining such results going to be a very daunting task.

Authors' response: Thank you for this comment. We have now clarified that the statistical assessment of bias will be in respect of the prevalence and statistical association studies, but not the qualitative studies reporting textual data. See "Data synthesis" (p.11).